

Week 1 — Fundamentals & Beta-Binomial**Bayes' theorem**

$$p(\boldsymbol{\theta}|\mathbf{y}) = \frac{p(\mathbf{y}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathbf{y})}$$

Marginal likelihood

$$p(\mathbf{y}) = \int p(\mathbf{y}|\boldsymbol{\theta})p(\boldsymbol{\theta})d\boldsymbol{\theta}$$

Beta distribution $\theta \sim \text{Beta}(a, b)$

$$p(\theta) = \frac{\theta^{a-1}(1-\theta)^{b-1}}{B(a, b)}, \quad B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

• Mean: $\frac{a}{a+b}$ Mode: $\frac{a-1}{a+b-2}$ Var: $\frac{ab}{(a+b)^2(a+b+1)}$ **Beta-Binomial conjugate update**

$$\begin{aligned} \text{Prior: } \theta &\sim \text{Beta}(a, b) \quad \text{Data: } y \sim \text{Bin}(N, \theta) \\ \Rightarrow \quad \theta|y &\sim \text{Beta}(a+y, b+N-y) \end{aligned}$$

Log marginal likelihood

$$\log p(y) = \log B(a+y, b+N-y) - \log B(a, b)$$

Week 2 — Grid Approx & Logistic Reg**Sigmoid / logistic**

$$\sigma(a) = \frac{1}{1+e^{-a}}, \quad \sigma'(a) = \sigma(a)(1-\sigma(a))$$

Binary logistic regression

$$p(y_n|w, x_n) = \sigma(w^\top x_n)^{y_n} (1 - \sigma(w^\top x_n))^{1-y_n}$$

Log joint (with Gaussian prior $w \sim \mathcal{N}(0, \alpha^{-1}I)$)

$$\log p(w|y) \propto \sum_n \log p(y_n|w, x_n) - \frac{\alpha}{2} \|w\|^2$$

Gradient & Hessian of log joint

$$\begin{aligned} \nabla_w \log p(w|y) &= X^\top (y - \mathbf{p}) - \alpha w \\ H &= -X^\top \text{diag}(\mathbf{p} \odot (1 - \mathbf{p}))X - \alpha I \end{aligned}$$

where $p_n = \sigma(w^\top x_n)$.**Grid normalisation**

$$\pi_{ij} = \frac{e^{\log p(\theta_i, \theta_j | \mathbf{y})}}{\sum_{k,l} e^{\log p(\theta_k, \theta_l | \mathbf{y})}}$$

Week 3 — Bayesian Linear Regression**Model**

$$p(\mathbf{y}|w, \Phi, \beta) = \mathcal{N}(\mathbf{y}|\Phi w, \beta^{-1}I), \quad p(w|\alpha) = \mathcal{N}(w|0, \alpha^{-1}I)$$

Posterior (exact, conjugate)

$$\begin{aligned} p(w|\mathbf{y}) &= \mathcal{N}(w|\mathbf{m}_N, S_N) \\ S_N &= (\alpha I + \beta \Phi^\top \Phi)^{-1} \\ \mathbf{m}_N &= \beta S_N \Phi^\top \mathbf{y} \end{aligned}$$

Posterior predictive

$$\begin{aligned} p(y^*|x^*, \mathbf{y}) &= \mathcal{N}(y^*|\mathbf{m}_N^\top \phi^*, \sigma_N^2(x^*)) \\ \sigma_N^2(x^*) &= \beta^{-1} + (\phi^*)^\top S_N \phi^* \end{aligned}$$

Log marginal likelihood

$$\begin{aligned} \log p(\mathbf{y}|\alpha, \beta) &= \frac{D}{2} \log \alpha + \frac{N}{2} \log \beta - \frac{N}{2} \log(2\pi) \\ &\quad - \frac{\beta}{2} \|\mathbf{y} - \Phi \mathbf{m}_N\|^2 - \frac{\alpha}{2} \mathbf{m}_N^\top \mathbf{m}_N + \frac{1}{2} \log |S_N| \end{aligned}$$

MLE (normal equations)

$$w_{\text{MLE}} = (\Phi^\top \Phi)^{-1} \Phi^\top \mathbf{y}$$

MAP

$$w_{\text{MAP}} = (\alpha I + \beta \Phi^\top \Phi)^{-1} \beta \Phi^\top \mathbf{y} = \mathbf{m}_N$$

Linear Gaussian systems — given $y = Ax + b + n$, $n \sim \mathcal{N}(0, \Sigma_n)$:

$$p(y|x) = \mathcal{N}(y|Ax + b, \Sigma_n)$$

*Joint then condition: $p(x, y)$ is joint Gaussian; use standard conditioning formulas.***Marginal / conditional of Gaussian**

$$\begin{aligned} p\begin{pmatrix} x \\ y \end{pmatrix} &= \mathcal{N}\left(\begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{pmatrix}\right) \\ p(x|y) &= \mathcal{N}(x|\mu_x + \Sigma_{xy}\Sigma_{yy}^{-1}(y - \mu_y), \Sigma_{xx} - \Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx}) \end{aligned}$$

Week 4 — Laplace Approximation

MAP estimation

$$w^* = \arg \max_w \log p(w|\mathbf{y})$$

Laplace approximation

$$q(w) = \mathcal{N}(w|w^*, \Lambda^{-1}), \quad \Lambda = -\nabla^2 \log p(w^*|\mathbf{y})$$

Posterior predictive approximations

Plug-in:

$$p(y^* = 1|x^*, \mathbf{y}) \approx \sigma(w^{*\top} x^*)$$

Probit approximation: ($\kappa^2 = \phi^{*\top} \Lambda^{-1} \phi^*$)

$$p(y^* = 1|x^*, \mathbf{y}) \approx \sigma\left(\frac{\mu_{f^*}}{\sqrt{1 + \pi\sigma_{f^*}^2/8}}\right)$$

where $\mu_{f^*} = w^{*\top} x^*$, $\sigma_{f^*}^2 = x^{*\top} \Lambda^{-1} x^*$.

Monte Carlo:

$$p(y^* = 1|x^*, \mathbf{y}) \approx \frac{1}{S} \sum_s \sigma(w^{(s)\top} x^*), \quad w^{(s)} \sim q(w)$$

Classification metrics

$$\text{Accuracy} = \frac{1}{N} \sum_n \mathbf{1}[\hat{y}_n = y_n]$$

$$\text{ELPD} = \frac{1}{N} \sum_n \log p(y_n|\mathbf{y}_{-n}, x_n) \approx \frac{1}{N} \sum_n \log \hat{p}_n^{y_n} (1 - \hat{p}_n)^{1-y_n}$$

Week 5 — Gaussian Process Regression

GP prior $f \sim \text{GP}(m(\cdot), k(\cdot, \cdot))$

$$\mathbf{f} \sim \mathcal{N}(0, K), \quad K_{ij} = k(x_i, x_j)$$

Kernels ($r = \|x - x'\|$, ℓ lengthscale, κ^2 magnitude)

SE: $\kappa^2 \exp\left(-\frac{r^2}{2\ell^2}\right)$

Matérn $\frac{1}{2}$: $\kappa^2 \exp\left(-\frac{r}{\ell}\right)$

Matérn $\frac{3}{2}$: $\kappa^2 \left(1 + \frac{\sqrt{3}r}{\ell}\right) \exp\left(-\frac{\sqrt{3}r}{\ell}\right)$

GP posterior (noise σ^2)

$$\mathbf{k}_* = [k(x^*, x_1), \dots, k(x^*, x_N)]^\top$$

$$\mu_{f^*} = \mathbf{k}_*^\top (K + \sigma^2 I)^{-1} \mathbf{y}$$

$$\sigma_{f^*}^2 = k(x^*, x^*) - \mathbf{k}_*^\top (K + \sigma^2 I)^{-1} \mathbf{k}_*$$

GP posterior predictive (with obs. noise)

$$p(y^*|\mathbf{y}, x^*) = \mathcal{N}(y^*|\mu_{f^*}, \sigma_{f^*}^2 + \sigma^2)$$

Log marginal likelihood

$$\log p(\mathbf{y}|\mathbf{x}, \kappa, \ell, \sigma) = -\frac{1}{2} \mathbf{y}^\top C^{-1} \mathbf{y} - \frac{1}{2} \log |C| - \frac{N}{2} \log(2\pi)$$

where $C = K + \sigma^2 I$.

Week 6 — GP Classification

Bernoulli likelihood

$$p(y_n|f_n) = \sigma(f_n)^{y_n} (1 - \sigma(f_n))^{1-y_n}$$

Laplace approximation on GP

$$q(\mathbf{f}) = \mathcal{N}(\mathbf{f}|\hat{\mathbf{f}}, (K^{-1} + W)^{-1})$$

where $\hat{\mathbf{f}} = \arg \max \log p(\mathbf{f}|\mathbf{y})$ and $W = -\nabla^2 \log p(\mathbf{y}|\mathbf{f})|_{\hat{\mathbf{f}}} = \text{diag}(\hat{p}_n(1 - \hat{p}_n))$.

Woodbury-stable form

$$(K^{-1} + W)^{-1} = K - KW^{1/2}(I + W^{1/2}KW^{1/2})^{-1}W^{1/2}K$$

Predictive posterior

$$\mu_{f^*} = \mathbf{k}_*^\top (K + W^{-1})^{-1} \mathbf{y}_{\text{eff}}, \quad \sigma_{f^*}^2 = k_{**} - \mathbf{k}_*^\top (K + W^{-1})^{-1} \mathbf{k}_*$$

Probit approximation (binary)

$$p(y^* = 1|\mathbf{y}, x^*) \approx \sigma\left(\frac{\mu_{f^*}}{\sqrt{1 + \pi\sigma_{f^*}^2/8}}\right)$$

Softmax

$$p(y = k|\mathbf{f}) = \frac{e^{f_k}}{\sum_j e^{f_j}}, \quad \mathbf{f} = W\phi(x)$$

Entropy

$$H[p] = - \sum_k p_k \log p_k \geq 0$$

Expected utility

$$a^* = \arg \max_a \sum_y p(y|x, \mathbf{D}) U(a, y)$$

For 0-1 loss $U(a, y) = \mathbf{1}[a = y]$: $a^* = \arg \max_k p(y = k|x, \mathbf{D})$.

Aleatoric vs. epistemic uncertainty

$$\mathbb{V}[y] = \underbrace{\mathbb{E}_\theta[\mathbb{V}[y|\theta]]}_{\text{aleatoric}} + \underbrace{\mathbb{V}_\theta[\mathbb{E}[y|\theta]]}_{\text{epistemic}}$$

ELBO

$$\mathcal{L}(q) = \mathbb{E}_q[\log p(\mathbf{y}, w)] - \mathbb{E}_q[\log q(w)] = \log p(\mathbf{y}) - \text{KL}[q||p]$$

Mean-field Gaussian $q(w) = \mathcal{N}(w|\mathbf{m}, \text{diag}(\mathbf{v}))$

Entropy:

$$H[q] = \frac{1}{2} \sum_d \log(2\pi e v_d)$$

ELBO gradient via reparameterisation $w = \mathbf{m} + \sqrt{\mathbf{v}} \odot \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$:

$$\nabla_\lambda \mathcal{L} = \nabla_\lambda \mathbb{E}_\epsilon[\log p(\mathbf{y}, \mathbf{m} + \sqrt{\mathbf{v}} \odot \epsilon)] + \nabla_\lambda H[q]$$

Analytical ELBO for linear Gaussian ($\mathbf{y} = Xw + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$, $w \sim \mathcal{N}(0, \kappa^2 I)$):

$$\begin{aligned} \mathcal{L} = & \underbrace{-\frac{1}{2\sigma^2} \|\mathbf{y} - X\mathbf{m}\|^2 - \frac{1}{2\sigma^2} \text{tr}(X \text{diag}(\mathbf{v}) X^\top)}_{\mathbb{E}[\log p(\mathbf{y}|\mathbf{w})]} \\ & + \underbrace{-\frac{1}{2\kappa^2} (\|\mathbf{m}\|^2 + \sum_d v_d)}_{\mathbb{E}[\log p(w)]} + H[q] \end{aligned}$$

KL between Gaussians

$$\text{KL}[\mathcal{N}(\mu_1, \Sigma_1) || \mathcal{N}(\mu_2, \Sigma_2)] = \frac{1}{2} \left[\log \frac{|\Sigma_2|}{|\Sigma_1|} - d + \text{tr}(\Sigma_2^{-1} \Sigma_1) + (\mu_1 - \mu_2)^\top \Sigma_2^{-1} (\mu_1 - \mu_2) \right]$$

Hamiltonian

$$H(\theta, \nu) = U(\theta) + K(\nu), \quad U(\theta) = -\log p(\theta|\mathbf{y}), \quad K(\nu) = \frac{1}{2} \nu^\top \nu$$

Leapfrog step (step size η , L steps):

$$\begin{aligned} \nu_{t+\frac{1}{2}} &= \nu_t - \frac{\eta}{2} \nabla U(\theta_t) \\ \theta_{t+1} &= \theta_t + \eta \nu_{t+\frac{1}{2}} \\ \nu_{t+1} &= \nu_{t+\frac{1}{2}} - \frac{\eta}{2} \nabla U(\theta_{t+1}) \end{aligned}$$

Accept/reject

$$\alpha = \min(1, \exp(H(\theta_t, \nu_t) - H(\theta^*, \nu^*)))$$

Gelman-Rubin $\hat{R}$ (want < 1.01):

$$\hat{R} = \sqrt{\frac{\hat{V}}{W}}, \quad \hat{V} = \frac{n-1}{n} W + \frac{B}{n}, \quad B = \frac{n}{m-1} \sum_j (\bar{\theta}_j - \bar{\theta})^2$$

Effective sample size

$$\text{ESS} = \frac{S}{1 + 2 \sum_{\tau=1}^{\infty} \rho_\tau}$$

where ρ_τ is the lag- τ autocorrelation.

Metropolis-Hastings (proposal $q(\theta'|\theta)$):

$$\alpha = \min\left(1, \frac{p(\theta'|\mathbf{y}) q(\theta|\theta')}{p(\theta|\mathbf{y}) q(\theta'|\theta)}\right)$$

Symmetric proposal (random walk): $\alpha = \min(1, p(\theta'|\mathbf{y})/p(\theta|\mathbf{y}))$.

Generative model

$$\begin{aligned}\pi &\sim \text{Dir}(\alpha_0 \mathbf{1}) \\ \Lambda_k &\sim \text{Wishart}(W_0, \nu_0) \\ \mu_k | \Lambda_k &\sim \mathcal{N}(\mathbf{m}_0, (\beta_0 \Lambda_k)^{-1}) \\ z_n | \pi &\sim \text{Cat}(\pi) \\ x_n | z_n, \mu, \Lambda &\sim \mathcal{N}(\mu_{z_n}, \Lambda_{z_n}^{-1})\end{aligned}$$

Variational family (mean-field)

$$q(Z, \pi, \mu, \Lambda) = q(Z) q(\pi) \prod_k q(\mu_k, \Lambda_k)$$

Responsibilities (E-step)

$$r_{nk} \propto \exp(\mathbb{E}[\log \pi_k] + \frac{1}{2} \mathbb{E}[\log |\Lambda_k|] - \frac{D}{2} \log(2\pi) - \frac{1}{2} \mathbb{E}_{\mu_k, \Lambda_k}[(x_n - \mu_k)^\top \Lambda_k (x_n - \mu_k)])$$

Dirichlet statistics

$$\mathbb{E}[\log \pi_k] = \psi(\alpha_k) - \psi\left(\sum_j \alpha_j\right), \quad \alpha_k = \alpha_0 + N_k, \quad N_k = \sum_n r_{nk}$$

Week 11 — Black-Box VI (BBVI)

Reparameterisation trick

$$\begin{aligned}w &= \mathbf{m} + \sqrt{\mathbf{v}} \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \\ \mathcal{L}(\lambda) &\approx \frac{1}{S} \sum_s \log p(\mathbf{y}, w^{(s)}) + H[q]\end{aligned}$$

Robust likelihood

$$p(y|w, x) = (1 - \epsilon_0) \text{Ber}(y|\sigma(w^\top x)) + \epsilon_0 \text{Ber}(y|\sigma(-w^\top x))$$

Probit likelihood

$$p(y = 1|w, x) = \Phi(w^\top x), \quad \Phi(t) = \int_{-\infty}^t \mathcal{N}(s|0, 1) ds$$

Week 12 — Bayesian Neural Networks

Architecture

$$h_l = \tanh(W_l h_{l-1} + b_l), \quad \hat{y} = W_L h_{L-1} + b_L$$

Gaussian likelihood (regression)

$$p(\mathbf{y}|w, X) = \mathcal{N}(\mathbf{y}|\hat{\mathbf{y}}(X, w), \sigma_n^2 I)$$

Log joint

$$\log p(w|\mathbf{y}) \propto \underbrace{-\frac{1}{2\sigma_n^2} \|\mathbf{y} - \hat{\mathbf{y}}(X, w)\|^2}_{\text{log-lik}} - \underbrace{\frac{1}{2\sigma_w^2} \|w\|^2}_{\text{log-prior}}$$

MAP training (=L2-regularised training): optimise $\log p(w|\mathbf{y})$ with Adam.

Key Distributions & Facts

Gaussian $\mathcal{N}(x|\mu, \sigma^2)$

$$\log p = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2}$$

$$H = \frac{1}{2} \log(2\pi e \sigma^2).$$

Multivariate Gaussian $\mathcal{N}(x|\mu, \Sigma)$

$$\log p = -\frac{D}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma| - \frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu)$$

Bernoulli $y \sim \text{Ber}(\theta)$: $\log p = y \log \theta + (1 - y) \log(1 - \theta)$.

Categorical $y \sim \text{Cat}(\pi)$: $p(y = k) = \pi_k$.

Poisson $y \sim \text{Pois}(\mu)$: $\log p = y \log \mu - \mu - \log y!$

Dirichlet $\pi \sim \text{Dir}(\alpha)$: $p(\pi) \propto \prod_k \pi_k^{\alpha_k - 1}$.

Woodbury identity

$$(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$$

Matrix det. lemma

$$|A + UCV| = |C^{-1} + VA^{-1}U| \cdot |C| \cdot |A|$$

Credibility interval: For $\theta \sim \mathcal{N}(\mu, \sigma^2)$, 95% CI $\approx \mu \pm 1.96\sigma$.

Posterior predictive (general)

$$p(y^*|\mathbf{y}) = \int p(y^*|\theta) p(\theta|\mathbf{y}) d\theta$$

Ancestral sampling: draw $\theta^{(s)} \sim p(\theta|\mathbf{y})$, then $y^{(s)} \sim p(y^*|\theta^{(s)})$.

Exam Quick Tips

- **Prior predictive:** no data yet, integrate prior over θ .
- **Posterior predictive:** use fitted posterior $p(\theta|\mathbf{y})$.
- **GP prior predictive:** $p(y^*|x^*) = \mathcal{N}(0, k(x^*, x^*) + \sigma^2)$.
- **Laplace is always around MAP:** find w^* first, then $H = -\nabla^2 \log p|_{w^*}$.
- **Probit vs. plug-in:** probit $\approx$ accounts for posterior uncertainty; plug-in ignores it.
- **ELBO $\leq$ log evidence:** equality iff $q = p(\cdot|\mathbf{y})$.
- **Mean-field KL:** $\text{KL}[q(w_1)q(w_2)||p(w_1, w_2)] \geq \text{KL}[q(w_1, w_2)||p(w_1, w_2)]$.
- **Higher α** (prior precision) $\Rightarrow$ weights shrink toward 0, less overfitting.
- **Softmax argmax** = argmax of f_k (no need to exponentiate).